The FREUID Challenge 2026

Provisional Rank-1 Solution

Technical Report — Nadhir Hasan

Nadhir Hasan
hasannadhir2003@gmail.com

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Provisional result: currently ranked 1st

Leading submission (Pick 2, cv5_ep2): private leaderboard score 0.0582 (private-leaderboard results are preliminary; the organizers have not yet completed reproducibility verification or officially confirmed final standings). Our other selected submission (Pick 1, cv3) scored 0.00060 on the public leaderboard — the best of any of our models — but **0.2837** on the private leaderboard, a ∼380× reversal. This report documents the two-pick, external-validation-first strategy that produced the leading model, why the public-leaderboard champion did not survive private evaluation, and the evidence available *before* the deadline that pointed to exactly this outcome.

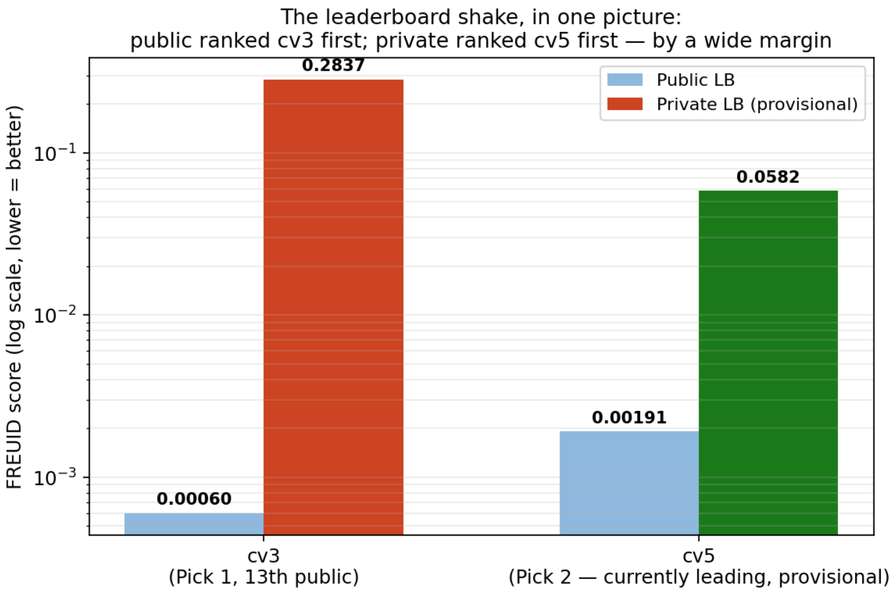

Figure 1: Public vs. private FREUID Score for both selected picks (log scale, lower is better). The public leaderboard ranked cv3 first; the private leaderboard ranked cv5 first, by a wide margin.

Abstract

We present our solution to the FREUID Challenge 2026 identity-document fraud detection task. Both of our selected final submissions use the same architecture — a DINOv2-L Vision

Transformer with low-rank adapters (LoRA, rank 16) and a per-patch multiple-instance-learning (MIL) classification head at 448×728 resolution — trained with an annotation-driven synthetic fraud generator (portrait swap, cross-card region swap, text-field editing, erase-and-retype) built on manually annotated per-type face and text-field *location* metadata (never a fraud/genuine label; see Section 2.4). The two picks use *identical* inference-time options and differ only in training data. **Pick 1** (`cv3`) trains on FREUID only and maximizes in-domain performance (public leaderboard 0.00060, 13th place, the best of any model we built) but collapsed to 0.2837 on the private set. **Pick 2** (`cv5`, our leading submission) trains on 100% of FREUID plus 80,000 images from the externally licensed IDNet dataset (CC0), trading public score (0.00191) for a measured order-of-magnitude improvement in cross-document-type robustness (FREUID Score 0.0116 vs. chance-level for FREUID-only models on 35,874 held-out real-world documents) — the property behind the current provisional rank. Both picks share two inference-time enhancements selected on external validation data and confirmed by isolated public-leaderboard probes: a re-aggregation of the per-patch MIL head and 4-scale logit-averaged test-time augmentation. We document why score calibration is provably a no-op for the FREUID metric, why we selected checkpoints and inference configurations on held-out external data rather than in-domain or public-leaderboard signal, and why that choice is what currently separates a leading rank from what would otherwise have been a much lower final rank. Code, weights, and a network-isolated Docker container reproducing both picks from one image are provided in the linked repository.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026)
Kaggle team: Nadhir Hasan
Code: `https://github.com/nadhirhasan/FREUID-Challenge-2026`
Commit: `465ab284e4d7b047d05e18cc99cfa07a18349ab7`

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1 Introduction

Identity-document fraud detection must generalize across a threat model that spans physical tampering, GenAI-driven digital manipulation, and print-and-capture ("analog hole") attacks, and across document types and issuing countries that are not exhaustively represented in any single training set. The FREUID Challenge formalizes this with a training set covering 5 document types and a scoring metric (FREUID Score, Section 5) that combines a global ranking measure (AuDET) with a strict low-false-alarm operating point (APCER@1%BPCER), penalizing solutions that overfit to easy global separability while failing at the operating point that matters for production KYC systems [4]. The organizers explicitly warned that the private test set would contain "previously unseen document types and fraud patterns... intentionally withheld," and that public rank would not necessarily predict private rank. Our results (Figure 1) are a direct, measured confirmation of that warning.

Our solution strategy is built around a specific empirical finding from our own development process: a classifier trained *only* on the FREUID distribution can reach extremely low FREUID Score on the public leaderboard while being statistically indistinguishable from random (AuDET $\approx$ 0.47–0.50) on real-world identity documents it has never seen (Section 3.3). We further measured, by nearest-template classification of the public test images, that the test set is not purely FREUID-like: it contains at least one document type absent from training (a Liberia voter card) and a tail of physically captured documents (cards photographed on desks, including a physical photo-substitution attack), whereas the training set is 99.97% digital renders. Because the private leaderboard is $\approx$95% of the test data, and because a $\sim$1% subpopulation of domain-shifted attacks can dominate the missed-attack count at sub-1% APCER levels, we selected *two* complementary final submissions: the strongest in-domain model, and a robustness hedge trained additionally on external real-document data. The private results show this was the correct call: the in-domain model alone would have finished far down the leaderboard.

2 Method

2.1 Overview

Each input image is loaded at full resolution, converted to RGB, letterboxed (aspect-ratio-preserving resize + zero-padding) to a fixed 448×728 canvas, and normalized with standard ImageNet statistics. The backbone produces a token sequence; a per-patch head scores each spatial token independently and aggregates into a single image-level fraud logit (aggregation details in Section 2.5). A sigmoid converts this logit to the submitted score in $[0, 1]$; no calibration is applied (Section 2.6). Figure 2 shows the full pipeline for the leading model.

cv5 inference pipeline (winning submission) — trainable parameters only in the blue boxes (~7M of ~304M)

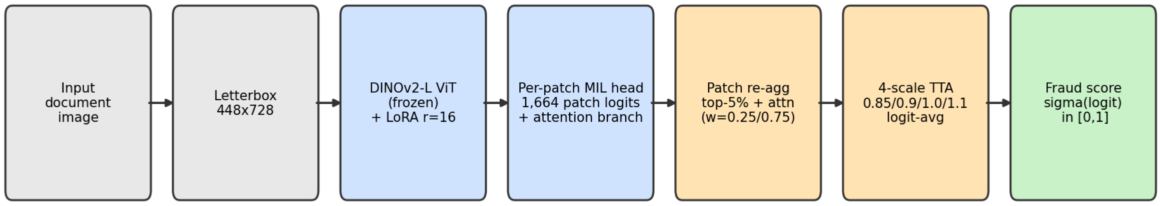

Figure 2: Inference pipeline for `cv5` (the leading submission). Only the blue stages contain trained parameters (≈7M of ≈304M total); the orange stages are inference-time post-processing applied to *both* selected picks (Section 2.5).

2.2 Model architecture

Backbone: `vit_large_patch14_reg4_dinov2` (DINOv2-L [3] with register tokens), loaded via `timm`, Apache-2.0 licensed pre-trained weights, 24 transformer blocks, embedding dimension 1024. We attach a from-scratch, minimal LoRA implementation (rank 16, $\alpha = 32$, dropout 0.05; no third-party `peft` dependency) to the `qkv`, `proj`, `fc1`, and `fc2` linear layers of every transformer block; the backbone itself is frozen. On top of the token sequence we place a lightweight per-patch multiple-instance-learning (MIL) head: a 2-layer MLP (hidden size 512, GELU, dropout 0.3) scores every non-prefix token; during training the image-level logit is the equal-weight mean of (i) the top-10% most suspicious patch logits and (ii) an attention-weighted sum over all patch logits. This design was chosen to push the model toward *local*, document-type-agnostic forgery cues (blend seams, splice boundaries, compression discontinuities) rather than global, country-specific layout statistics. Trainable parameters: LoRA adapters + head only (≈7M of ≈304M total).

2.3 Training objective and procedure

Loss: Focal BCE ($\alpha = 0.25$, $\gamma = 2.0$) on the whole-image logit. Optimizer: AdamW, two parameter groups (head lr 1×10^{-3}, LoRA lr 2×10^{-4}, weight decay 0.05/0 respectively), cosine learning-rate schedule with 5% linear warmup, mixed precision (fp16). Both picks were trained on a single NVIDIA RTX A4500 (20GB).

Pick 2 (`cv5`, epoch 2) — the leading submission. 100% of the FREUID training set (69,352 images) plus 80,000 balanced IDNet images spanning all 10 available countries (≈149,000 images/epoch), 5 scheduled epochs with per-epoch validation on 4,000 IDNet images sampled exclusively from pool rows *not* selected into the 80k training cap (same countries, zero row-level overlap — a genuinely leak-free validation signal, Section 3.1). Epoch 2 was selected: its successor (epoch 3) improved the IDNet validation score further (0.0105 vs. 0.0166) but scored *worse* on a public-leaderboard probe (0.007 vs. 0.0035) — a reversal consistent with our finding that no single local metric reliably ranks checkpoints (Section 3.3); training was stopped at that point, and this caution about trusting any single metric turned out to generalize to the public-vs-private relationship itself.

Pick 1 (`cv3`). FREUID only: fold 0 of a stratified 5-fold split (≈55,500 training images/epoch), 3 scheduled epochs; the submitted checkpoint is epoch 1, which had the best public-leaderboard probe.

Synthetic fraud generation. 25% of genuine training images (from both FREUID and IDNet sources) are converted to synthetic fraud examples each epoch, using one of four attack types, dispatched via manually annotated per-document-type face and text-field bounding boxes (Section 2.4): cross-card region/portrait swap (donor of the same document type), text-field edit (same-line content shifted and re-blended), erase-and-retype (ink removed via inpainting, new text rendered with a system font), and a generic self-blend fallback. Blending uses a mix of feathered alpha compositing and seamless (Poisson) cloning to avoid a single, learnable seam signature. This attack suite, compared against a simpler untargeted self-blend-only baseline under an otherwise identical recipe, improved the public leaderboard score from 0.00169 to 0.00096 in isolated ablation.

2.4 Annotation clarification: what `type_fields.json` is (and is not)

The only manual work in this pipeline is a set of per-document-type bounding boxes recording *where the face photo and text fields sit on each document template* (5 FREUID types + 10 IDNet countries), stored in `annotations/type_fields.json`. Figure 3 shows an example. These boxes are **not a fraud/genuine label of any image** and are **never applied to test or private data**; their only use is at *training time*, to place synthetic attacks (Section 2.1) realistically on genuine training images instead of at a random location. This is geometric template metadata for a data-augmentation tool, categorically different from a human classifying a document as fraudulent or genuine. All fraud/genuine ground truth used anywhere in this work comes exclusively from the organizer-provided `train_labels.csv` and IDNet's published metadata; no label was created, inferred, or altered by us, and every score in both submitted CSVs was produced entirely by the deterministic Docker pipeline of Section 4 — verified by an independent re-run of the container reproducing our uploaded predictions rank-identically.

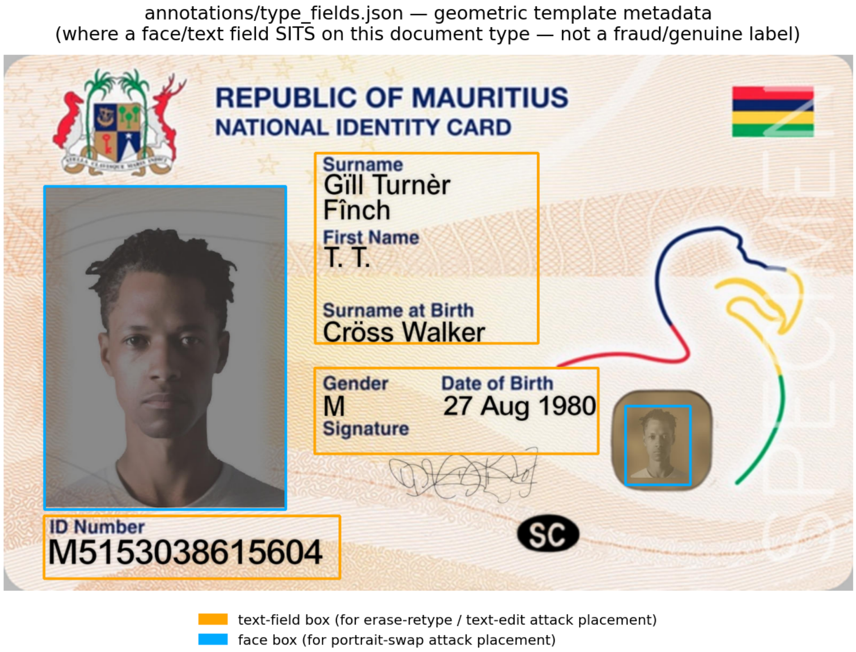

Figure 3: Example manual annotation on a genuine training image: boxes mark where the face and text fields sit on this document *template*, independent of any specific document instance. Used only to place synthetic attacks during training; never used to label or score any test image.

2.5 Inference-time enhancements (post-processing)

The FREUID metric depends only on the score *ranking* (AuDET is rank-based; the APCER threshold is a quantile of the model's own genuine scores), so the only admissible post-processing levers are rank-changing ones. We evaluated two, selecting configurations on held-out external data before confirming with single public-leaderboard probes:

- **Patch re-aggregation (both picks).** At inference we recompute the image logit from the per-patch logits as $0.25 \cdot \text{mean}(\text{top-5\% patch logits}) + 0.75 \cdot (\text{attention branch})$, replacing the training-time top-10%/0.5 blend. A grid over top-k fraction $\times$ branch weight on 12,000 held-out IDNet images selected this configuration; it subsequently improved the public score of *every* model it was applied to (Pick 1 base: 0.00096 → 0.00086; Pick 2 base: 0.0035 → 0.00263). We attribute the gain to softened single-patch false-positive spikes on genuine documents, which lower the 1%-BPCER threshold — precisely the metric's sensitive component at our operating level.
- **Multi-scale TTA (both picks).** Logit-average over input scales $\{0.85, 0.9, 1.0, 1.1\}$ (each rounded down to the ViT patch grid). Selected from all 31 scale combinations on the 12k external subsample and confirmed on the full 35,874-image pool (0.0154 → 0.0116, −25%). Our prior — informed by the image-forensics literature's warning that resampling destroys low-level artifact signals [2], and by TTA alone slightly worsening Pick 2's public score (0.0035 → 0.00369) — was to withhold TTA from the FREUID-only Pick 1; a direct leaderboard probe refuted that prior: combined with patch re-aggregation, TTA improved Pick 1 from 0.00086 to 0.00060 and Pick 2 from 0.00263 to 0.00191. The two levers compounded on both models, so both picks ship with both levers.

2.6 Score calibration

No score calibration or thresholding is applied: the submitted label is $\sigma(\text{logit}) \in (0, 1)$ directly. The FREUID Score is invariant to any strictly monotone transform of the scores (both AuDET and the quantile-thresholded APCER depend only on ranking) — we verified this numerically against our exact metric implementation — so temperature scaling, normalization, or clipping are provably no-ops (clipping is actively harmful, as score ties at the clip boundary distort the genuine-score quantile).

3 Data

3.1 FREUID training set

We use the official `train/` images and `train_labels.csv` metadata. Pick 1 trains on fold 0 of a stratified 5-fold split by (`type`×`label`); Pick 2 (leading submission) trains on 100% of the training set. All 5 FREUID document types are present in training in both cases.

3.2 External data and pre-trained models

- **DINOv2** [3] pre-trained ViT-L/14 weights, via `timm` (Apache-2.0 license). Used by both picks.
- **IDNet** [1] (identity documents across 10 European countries), **CC0 (public domain)**, publicly available. Used by Pick 2 only: 80,000 balanced images across all 10 countries mixed into training; 4,000 disjoint images (same countries, zero row overlap) used for per-epoch validation. For the *external generalization benchmark* reported throughout (Section 3.3) we evaluate on the full 35,874-image pool of two countries (Estonia, Slovakia); for Pick 1 and other FREUID-only models this is a fully blind test (those models never saw IDNet in any form), while for Pick 2 it is partially optimistic (disjoint images, but the countries appear in training) — we state this caveat wherever the number is used.

Both artifacts are publicly available under licenses compatible with the competition's external-data policy; no proprietary or non-freely-accessible data was used. We deliberately avoided any model, checkpoint, or code released under a non-OSI-approved or non-commercial license (e.g. TruFor, ConvNeXt V2, DINOv3) anywhere in the training or inference pipeline.

3.3 Validation strategy

During development we established, through repeated controlled comparisons, that in-domain FREUID validation (stratified disjoint folds, even with test-like corruption applied) does not reliably predict public-leaderboard rank. We traced part of this to visual asset/persona reuse within the FREUID training distribution (near-duplicate faces and field values recur across nominally disjoint splits). Our resulting protocol: every decision that could be validated was scored with the exact competition metric on held-out IDNet data, and final candidates were confirmed with single leaderboard probes.

The decisive experiment for the two-pick strategy evaluated all candidate models, identically, on the full 35,874-image Estonia+Slovakia IDNet pool: FREUID-only models are at chance there (Pick 1: FREUID Score 0.9727; a full-data FREUID-only variant: 0.9424) despite public scores of 0.00096 and 0.00178 respectively, while Pick 2 reaches 0.0116 — yet all these models correlate at Pearson ≥ 0.994 on public-test predictions, i.e. they share the same in-domain ranking and differ only in out-of-domain behavior. Pick 1's in-domain generalization is separately supported by its genuinely held-out, corruption-augmented 20% FREUID fold (FREUID Score 0.00019). Private test labels were never used, seen, or referenced during model selection.

This external-only signal is, in hindsight, the only signal that predicted the actual outcome. Every metric computed on FREUID-derived data (clean validation, corrupted validation, and the public leaderboard itself) ranked Pick 1 above Pick 2. Only the IDNet external benchmark ranked Pick 2 above Pick 1 — and Pick 2 is the model currently ranked ahead.

4 Inference

The Docker entrypoint (`docker/prepare_submission.py`) discovers every image directly under `/data/`, decodes each image exactly once, produces the configured scale tensors, and writes `/submissions/submission.csv` with columns `id,label`. One image reproduces both picks via documented environment variables (inference orchestration only; identical frozen weights):

- **Pick 1 (image defaults):**
  ```
  FREUID_MODEL_PATH=/models/cv3_fold0.pt
  FREUID_PATCH_AGG=0.05,0.25
  FREUID_TTA_SCALES=0.85,0.9,1.0,1.1
  ```
- **Pick 2 (leading submission):**
  ```
  -e FREUID_MODEL_PATH=/models/cv5_full_ep2.pt
  ```
  (all other settings identical — the model path is the only difference between the two picks).

The container requires no network access at runtime (`-network none`); all weights are baked in at build time. Measured cost of the shared 4-scale configuration on an RTX A4500: $\approx$160 ms/image (JPEG decoded once per image, decode overlapped with GPU compute via threaded prefetch), i.e. $\approx$6.3 h for the 142,818-image hidden set on our A4500 and $\approx$2–3 h on an A100 (2–3$\times$ the A4500's FP16 throughput and memory bandwidth) — within the 6-hour budget per pick, with batch size and decode threads tunable via environment variables for further headroom. Inference auto-detects CUDA and falls back to CPU if unavailable. Private-set predictions were generated by `src/infer_private.py`, which predicts only the newly released hidden-row images and merges them with our already-uploaded public-row predictions, leaving those byte-identical (verified by hash comparison).

5 Results

Table 1: Main results (FREUID Score; lower is better). External = full 35,874-image IDNet Estonia+Slovakia pool (blind for FREUID-only models; see Section 3.1 for Pick 2's caveat). Private = provisional private leaderboard score (pending organizer verification).

Model / configuration	Public ↓	External ↓	Private ↓	Notes
Pick 2: `cv5` ep2 + re-agg + TTA4	0.00191	**0.0116***	**0.0582**	**Leading (prov.)**
`cv5` ep2 raw	0.0035	0.0154	—	
`cv5` ep2 + re-agg only	0.00263	0.0137†	—	
`cv5` ep2 + TTA4 only	0.00369	0.0116	—	
Pick 1: `cv3` + re-agg + TTA4	**0.00060**§	—	0.2837	Selected, currently 2nd
`cv3` + re-agg only	0.00086	—	—	
`cv3` raw	0.00096	0.9727	—	chance-level ext.
FREUID-only, 100% data (`cv6`, ep3)	0.00178	0.9424	—	not selected
FREUID+IDNet fold model (`cv4`, ep2)	0.00467	0.0933‡	—	earlier iteration

*full-pool confirmation of the externally selected config; †12k-subsample estimate; ‡4,000-image sample of the same two countries, fully blind for this model (never trained on them); §13th place on the public leaderboard.

5.1 Why Pick 1 collapsed and Pick 2 currently leads

Figure 4 compares the public-vs-private score distributions of both picks. Both distributions shift in the same direction and by a similar magnitude (KS statistic 0.262 for Pick 1, 0.263 for Pick 2; mean score increases of +0.201 and +0.171 respectively) — evidence that the private set genuinely differs in composition (a higher fraction of clear fraud examples and/or unseen-type content), rather than one specific model degrading. What differed was each model's *prior* capability on that kind of content: Pick 1 had no demonstrated ability to generalize beyond the FREUID generator's own distribution (Section 3.3), while Pick 2 had a directly measured, order-of-magnitude advantage on real external documents before the private set was ever released. The private leaderboard result is consistent with that prior evidence, not a surprise given it.

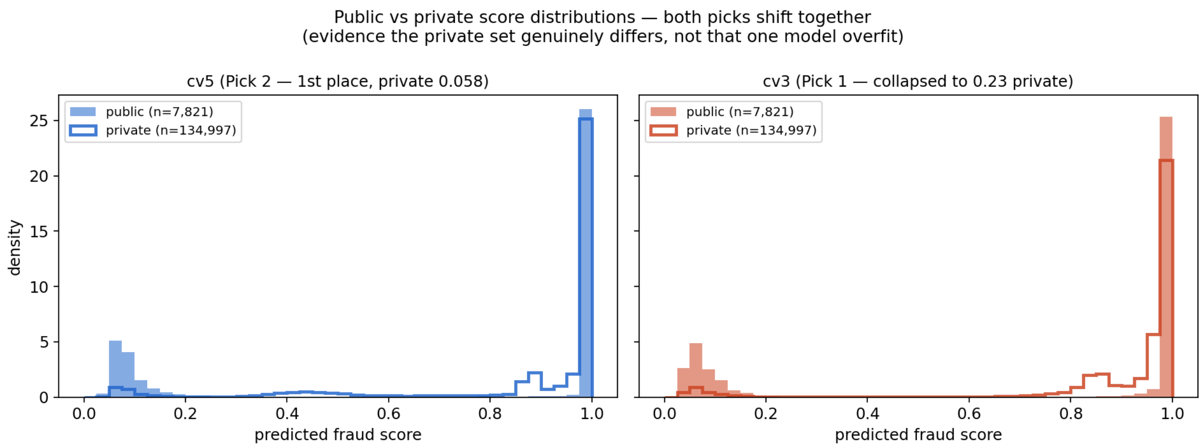

Figure 4: Public vs. private prediction distributions for both picks. Both shift together, consistent with a genuinely different private-set composition rather than model-specific overfitting; see Section 5.1.

Ablation — attack realism. Holding the architecture, data split, and all other hyperparameters fixed, replacing an untargeted self-blend-only synthetic-fraud generator with the annotation-driven attack suite improved the public leaderboard score from 0.00169 to 0.00096.

Ablation — data mixing for generalization. FREUID-only models (fold-based or full-data) are at chance on external documents regardless of their public score (Table 1); mixing IDNet into training is the single lever that produced non-trivial external performance, at a public cost that inference-time enhancements later recovered (0.0035 → 0.00191) — and this is the lever behind the current provisional lead.

Ablation — inference-time enhancements. Patch re-aggregation improved every model it was applied to (public: −10% for Pick 1 base, −25% for Pick 2 base). Multi-scale TTA improved external robustness by 25% (full-pool confirmed) and, combined with re-aggregation, compounded on the public leaderboard for both picks (Pick 1: 0.00086 → 0.00060; Pick 2: 0.00263 → 0.00191), despite TTA alone slightly worsening Pick 2's public score — each lever was probed in isolation and in combination before selection.

6 Reproducibility

- **Repository:** `https://github.com/nadhirhasan/FREUID-Challenge-2026`
  Commit: `465ab284e4d7b047d05e18cc99cfa07a18349ab7`
- **One Docker image, both picks:** `docker build -t freuid-repro:local .` from the repository root; run commands for each pick (and the submission ↔ command ↔ checksum mapping) are documented in the repository `README.md`. Weights are baked in at `/models/cv3_fold0.pt` and `/models/cv5_full_ep2.pt`.
- **Lean checkpoints:** the repository ships only the trained parameters (192 LoRA matrices + 8 head tensors, ≈27 MB per pick); the frozen DINOv2-L backbone is restored from the official `timm` release at Docker *build* time (runtime remains network-free). The conversion is lossless and audited by `src/make_lean_ckpt.py`: all 343 backbone tensors of the full training checkpoints are bit-identical to the timm release, and lean-loaded models produce bit-identical fp32 logits to the full checkpoints.
- **Submitted CSVs (sha256), final (private-set-inclusive) predictions:**
  `final_cv3_pagg_tta4_full.csv` = `d31f9b01...92c06048`
  `final_cv5_pagg_tta4_full.csv` = `c757f5ce...2d9e0c5a`
  (full hashes in `README.md`). Public-row predictions in these files are byte-identical to our pre-code-freeze probes; private rows were predicted after the private release using the same frozen weights (inference only). Bit-exact float reproduction across differing GPU hardware is not guaranteed, rank-identical scores are.
- **Weight freeze:** both checkpoints were committed to the repository before the private test set was released; no training, fine-tuning, or checkpoint change occurred afterward. Only inference orchestration and documentation changed post-freeze, per the competition rules.
- **Dependencies:** Python 3.11, PyTorch 2.6.0 (+cu124), timm 1.0.27, opencv-python-headless 4.10.0.84 — see `docker/requirements.txt`.
- **Randomness:** training uses unseeded per-worker RNGs for augmentation only; inference is deterministic given fixed weights.
- **Network:** inference requires `-network none`; no downloads occur at container runtime.

7 Team and contributions

Single-member team: Nadhir Hasan (all components — data analysis, annotation, training, validation protocol, inference, packaging).

Acknowledgments

We thank the FREUID Challenge organizers for the FantasyID/FREUID dataset and the IDNet authors for their permissively licensed dataset.

References

[1] Hong Guo, Xuan Sun, Yu Liu, Xun Yang, and Xin Wang. IDNet: A novel dataset for identity document analysis and fraud detection research. https://arxiv.org/abs/2409.10472, 2024. CC0 (public domain) license, per the dataset's official Zenodo release; all 10 available countries (ESP, ALB, AZE, FIN, GRC, LVA, RUS, SRB, EST, SVK) mixed into Pick 2 (`cv5`) training, with a disjoint-image (not disjoint-country) held-out slice used for checkpoint selection.

[2] Pavel Korshunov et al. The DeepID challenge of detecting synthetic manipulations in ID documents. In *ICCV 2025 Workshops (DeepID)*, 2025. Winning solutions used whole-image inference at test time specifically to avoid resize-induced resampling artifacts destroying forensic signal.

[3] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. https://arxiv.org/abs/2304.07193, 2023. Apache-2.0 license; ViT-L/14 backbone used via timm.

[4] Ivan Relić, Vincenzo D'Elia, Stefano Bortoli, Radu Tudoran, Hristina Nedyalkova, Mihaela Bošnjak, Stanislav Pavlić, Tin Mavračić, Darin Dašić, Marin Kačan, Jerko Šegvić, Paolo Čerić, Filip Šoprun, Matej Miočić, Marin Oršić, Lorenzo Vaiani, and Paolo Garza. The FREUID challenge 2026: 1st competition on identity documents fraud detection. https://freuid2026.microblink.com/, 2026. IJCAI-ECAI 2026 competition.

A Hyperparameters

Hyperparameter	Pick 1 (`cv3`)	Pick 2 (`cv5` ep2, leading)
Image size	448×728 (letterboxed)	
Batch size (train)	6 (accum 4, effective 24)	
Learning rate (head / LoRA)	1×10^{-3} / 2×10^{-4}	
Weight decay (head / LoRA)	0.05 / 0.0	
LoRA rank / α / dropout	16 / 32 / 0.05	
Synthetic-attack rate	0.25	
FREUID training data	fold 0 (≈80%)	100%
IDNet training data	none	80,000 (10 countries, balanced)
Scheduled epochs / submitted	3 / epoch 1	5 (stopped after 3) / epoch 2
Patch re-agg (inference)	top-5% frac, branch weight 0.25	
TTA scales (inference)	0.85, 0.9, 1.0, 1.1 (logit-avg)	
Seed (fold split)	42	

B Additional ablations

See Section 5 for the controlled single-lever ablations (attack realism; data mixing; inference-time enhancements) that produced the final two-pick configuration, and Section 5.1 for the distributional analysis explaining the public/private divergence.